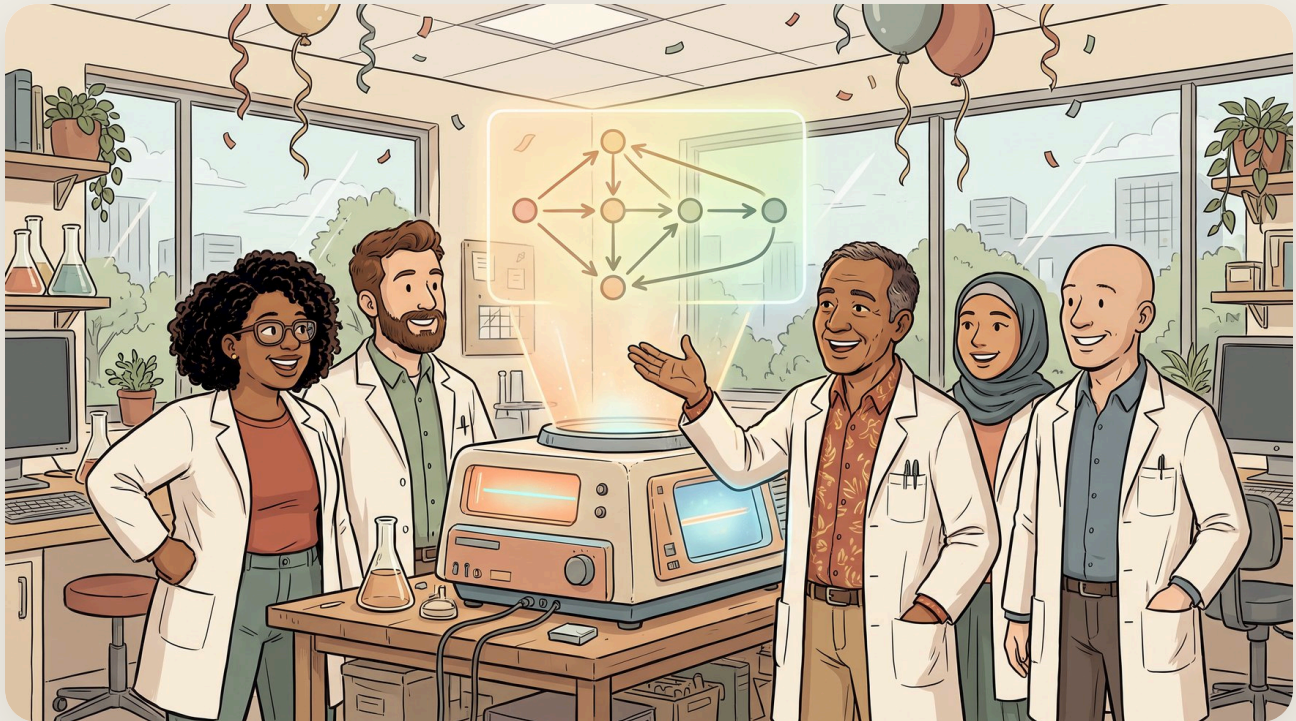

AYASS BIOSCIENCE

BiRAGAS

The Deployed Causal Engine Field Guide

An Illustrated Guide · How the Engine Was Built and How to Run It · 10 Questions



Inside the runnable BiRAGAS causal engine: how ten evidence modules wire into one machine that discovers cause from correlation, simulates interventions, refutes and abstains, and issues a per-target causal verdict, with 34 tests and a benchmark to prove it.

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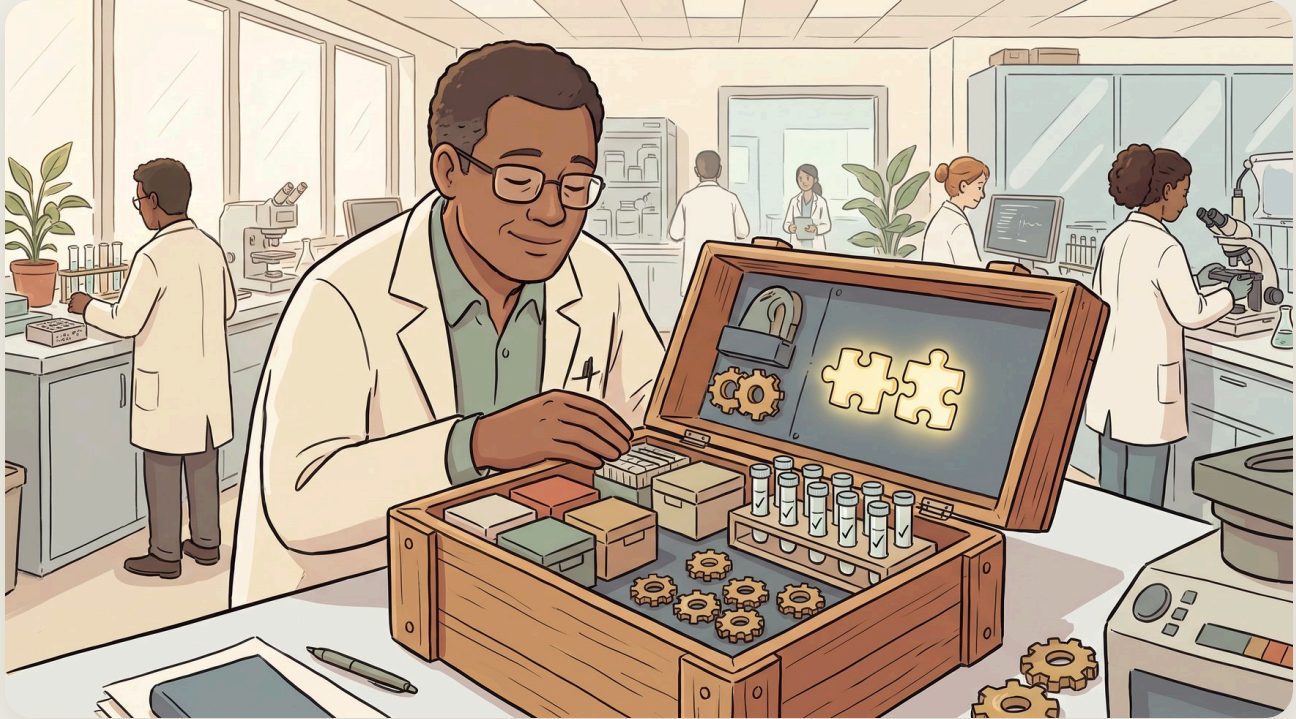
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THE PACKAGE

Q1 What exactly is the deployed causal engine?

**HOW BIRAGAS HELPS**

A runnable, integrated engine that turns BiRAGAS discovery from correlation-seeded into genuinely causal. It ships as engine code, real adapters for every module, verified integration patches, a JSON config, 34 automated tests, a benchmark, and a deployment guide. It replaces the $|r| > 0.50$ seeding and the hand-set 0.70 weight, keeps the excellent AIPW estimator, and adds a structural model, refutation and honest abstention.

WHAT IT IS

A runnable causal engine

KEEPS

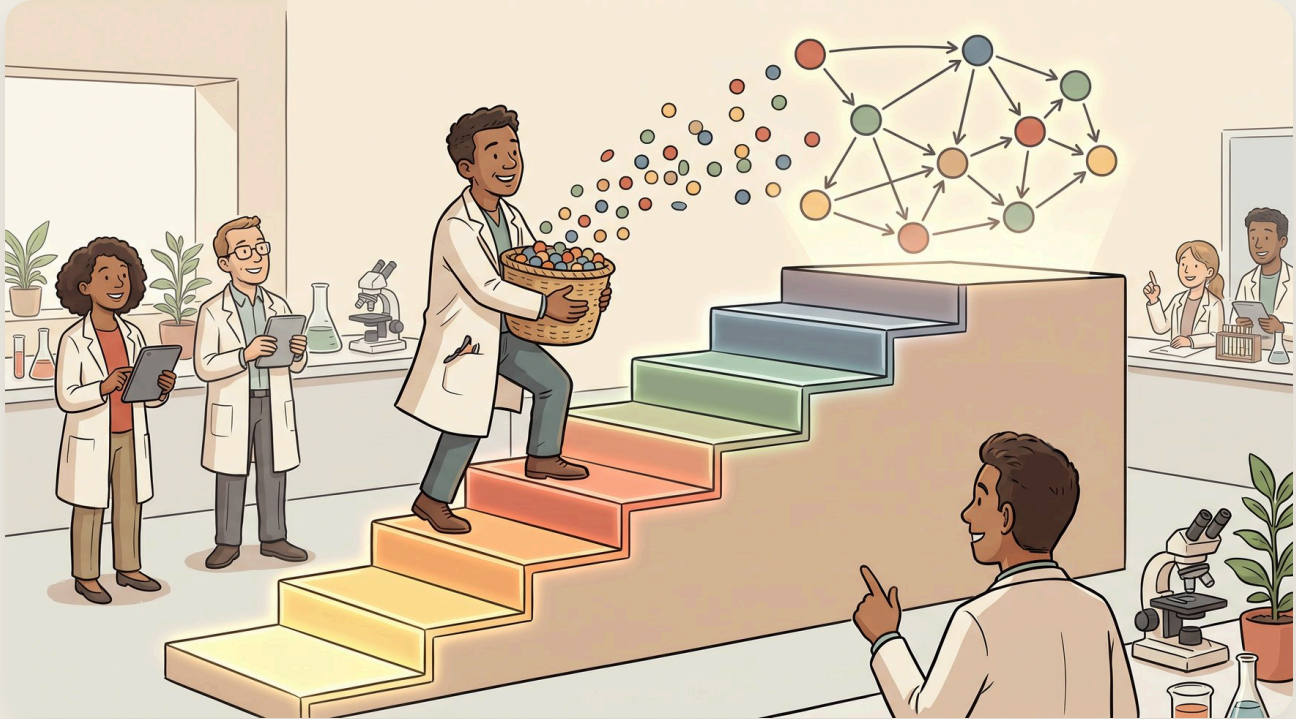
The AIPW estimator

REPLACES

Correlation seeding

SHIPS WITH

34 tests, benchmark, patches

THE ARCHITECTURE**Q2** What are the seven stages, end to end?**HOW BIRAGAS HELPS**

Ingest every source onto one scale; build a conditional-independence skeleton; run the discovery ensemble over bootstraps for stability; anchor and orient with interventional data; prune with CI, FCI and stability; fuse into a calibrated per-edge posterior; then estimate, refute and abstain to a per-edge verdict. Scattered signals enter; a certified causal DAG comes out.

STAGES

7 (ingest to verdict)

EXIT

A certified causal DAG

ENTRY

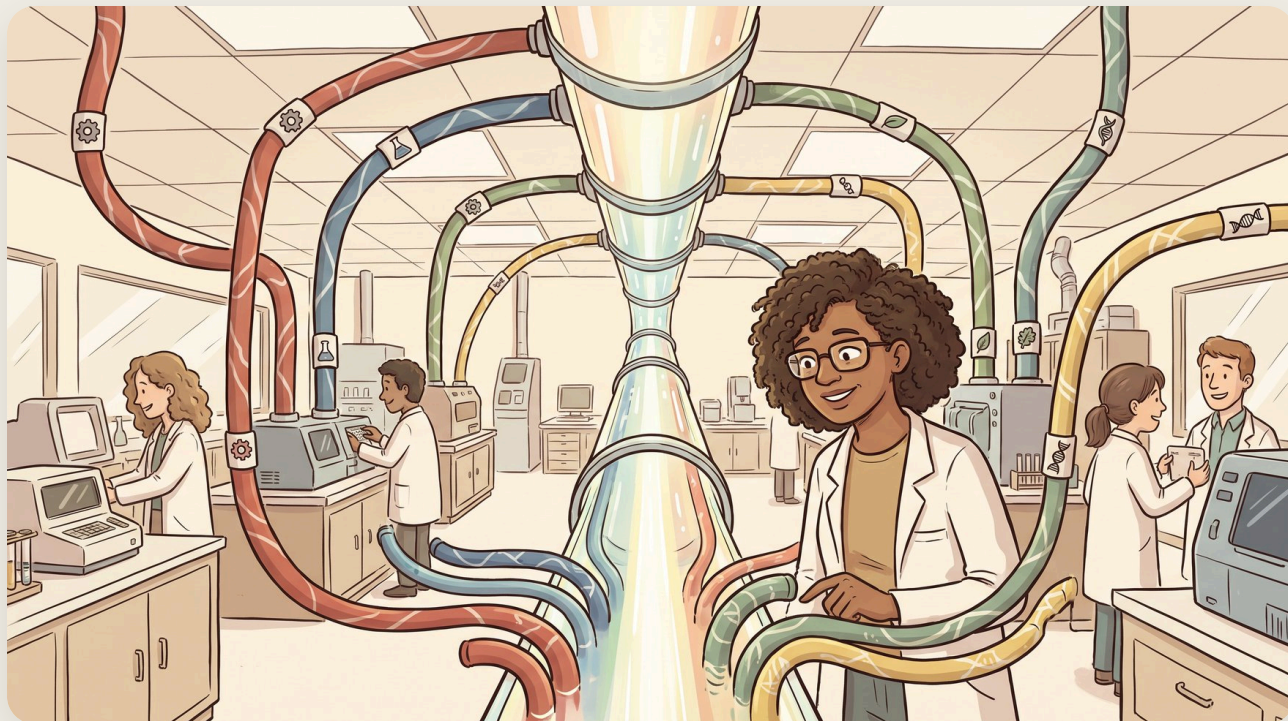
Ten evidence sources

UNCHANGED

The gold-standard estimator

THE EVIDENCE BUS

Q3 How do ten modules feed one engine?



HOW BIRAGAS HELPS

One assembler reads all ten from named data keys, each on a common scale. Seven are edge-level (CRISPR-IV, L1000, MR, SIGNOR, DoRothEA, Granger, ensemble); three are node-level driver signals (DepMap, MAGECK, PROGENy). Populate any subset; missing sources degrade gracefully and are never fabricated.

EDGE MODULES

7 sources

CONTRACT

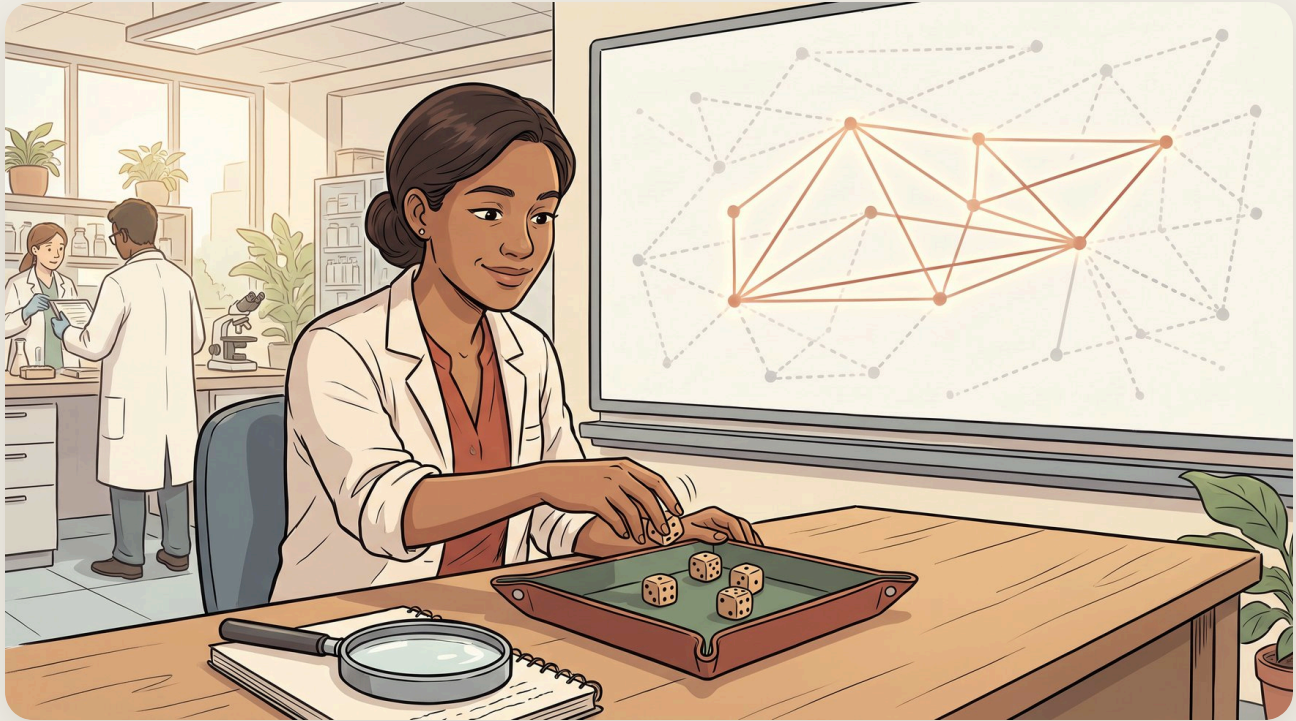
One data key each

NODE MODULES

3 driver signals

RULE

Missing degrades, never fabricated

DISCOVERY**Q4** How does it find real causal edges, not correlations?**HOW BIRAGAS HELPS**

Membership becomes a conditional-independence question, so a correlation running entirely through a confounder is removed. Then bootstrap stability selection keeps only the edges that recur across resamples, with a finite false-edge bound, replacing the guessed 0.70 weight with a guarantee. A kernel test option handles nonlinear signal.

SKELETON

CI test: Fisher-Z or KCI

REPLACES

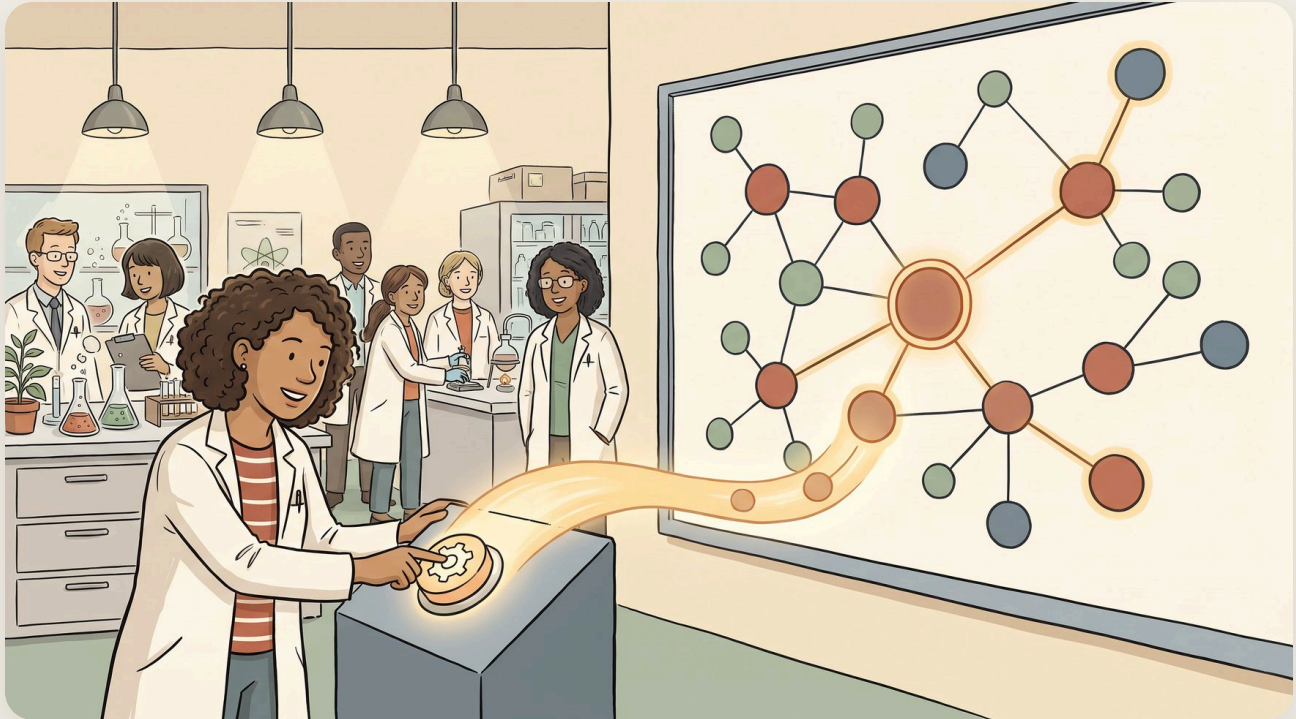
$|r| > 0.50$ and the $x0.70$ weight

STABILITY

Bootstrap, bounded false edges

RESULT

Confounded edges pruned

INTERVENTION**Q5** How does it simulate inhibiting a target?**HOW BIRAGAS HELPS**

A structural causal model is fit on the certified graph, one equation per gene from its parents, so the engine answers $\text{do}(X=x)$: clamp the target, propagate through the fitted equations, and read the effect on the disease program, with counterfactuals for individual profiles. This replaces the old linear path-flow that could never express a real intervention.

MODEL

Fitted structural causal model

ALSO

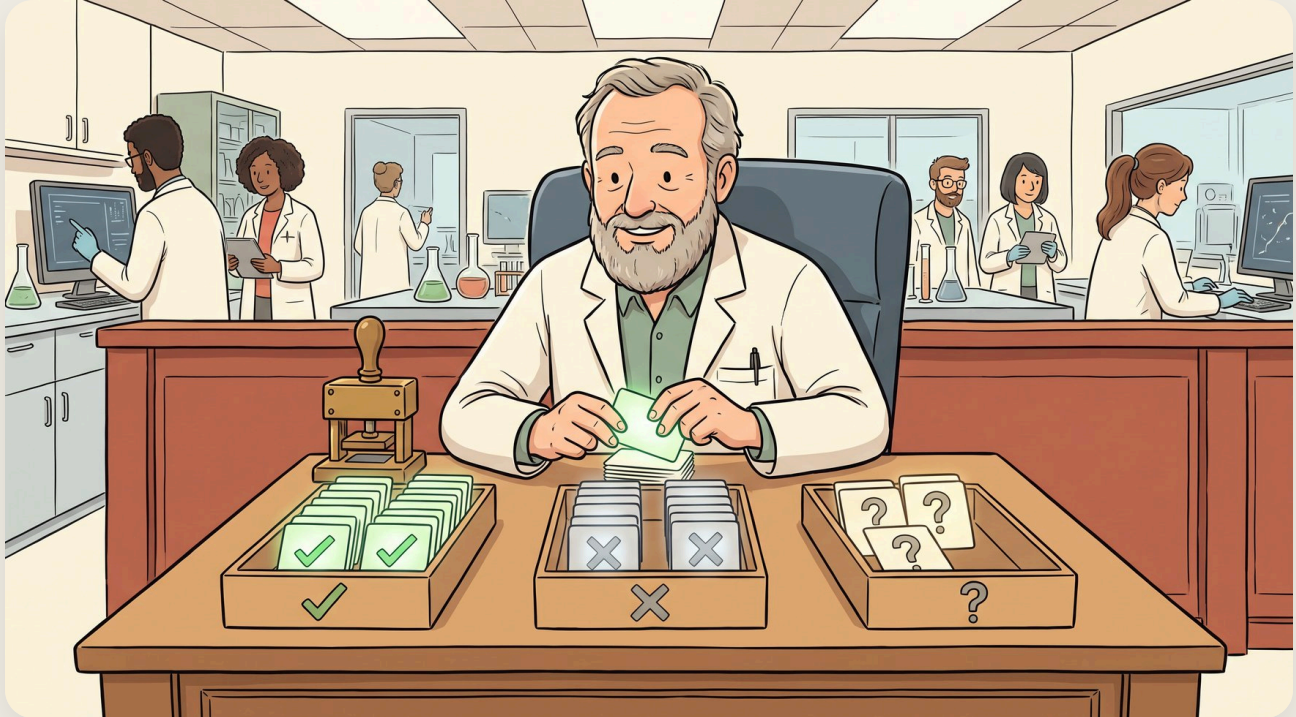
Counterfactuals per profile

OPERATOR

$\text{do}(\text{inhibit this target})$

REPLACES

Linear Phase-4 flow

THE VERDICT**Q6** How does it refute claims and stay honest?**HOW BIRAGAS HELPS**

Every effect is stress-tested with a placebo (permute the treatment and a real effect vanishes) and bootstrap sign-stability (a stable sign with a confidence interval excluding zero). Edges under latent confounding are declared not identified rather than reported. The pipeline closes with an honest per-edge verdict.

TESTS

Placebo + bootstrap stability

VERDICT

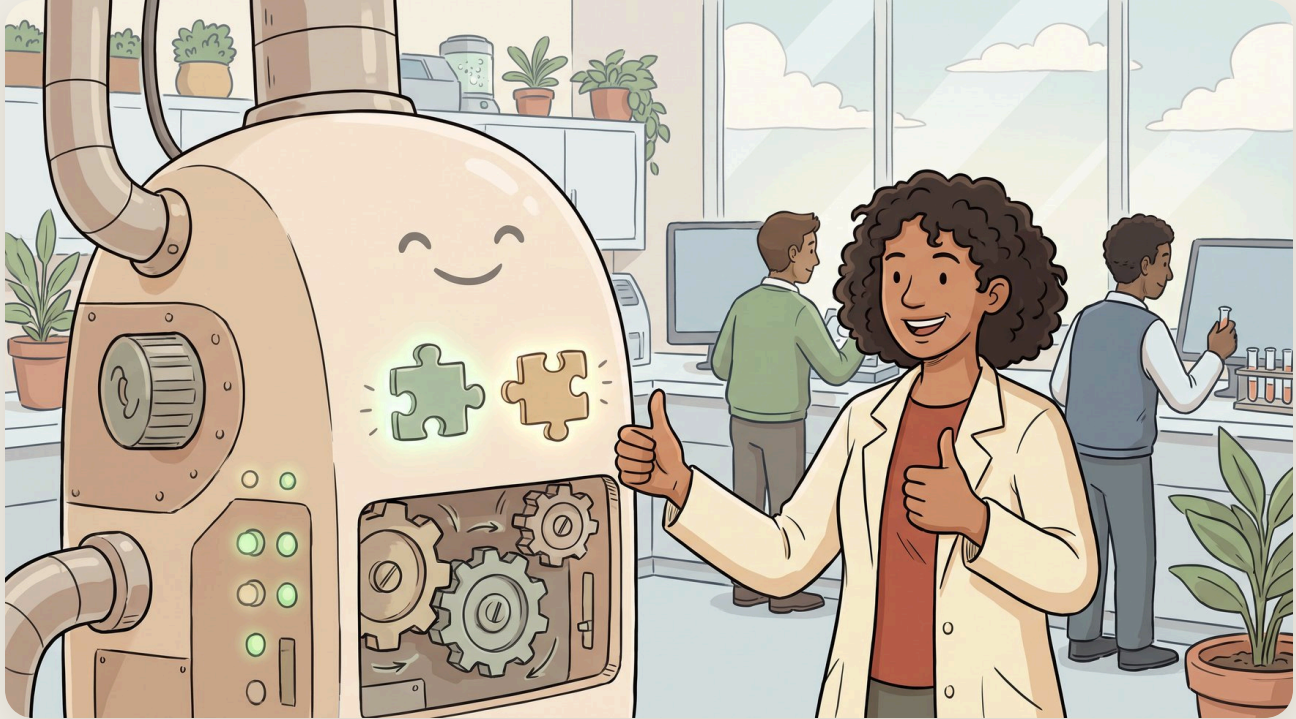
CAUSAL / REFUTED / NOT_IDENTIFIED

ABSTAINS ON

Latent confounding

STANDARD

Report only what survives

DEPLOYMENT**Q7** How does it wire into the live platform?**HOW BIRAGAS HELPS**

Two small, verified patches make the live graph builder decide membership by conditional independence and re-score by the fusion posterior; the heavy logic lives in a new module, so the production diff stays tiny and reversible behind config flags. The patches are generated from, and dry-run-applied to, the real source. A re-wiring, not a rewrite.

CHANGE

2 minimal, verified patches

PROOF

Dry-run apply to real source

RISK

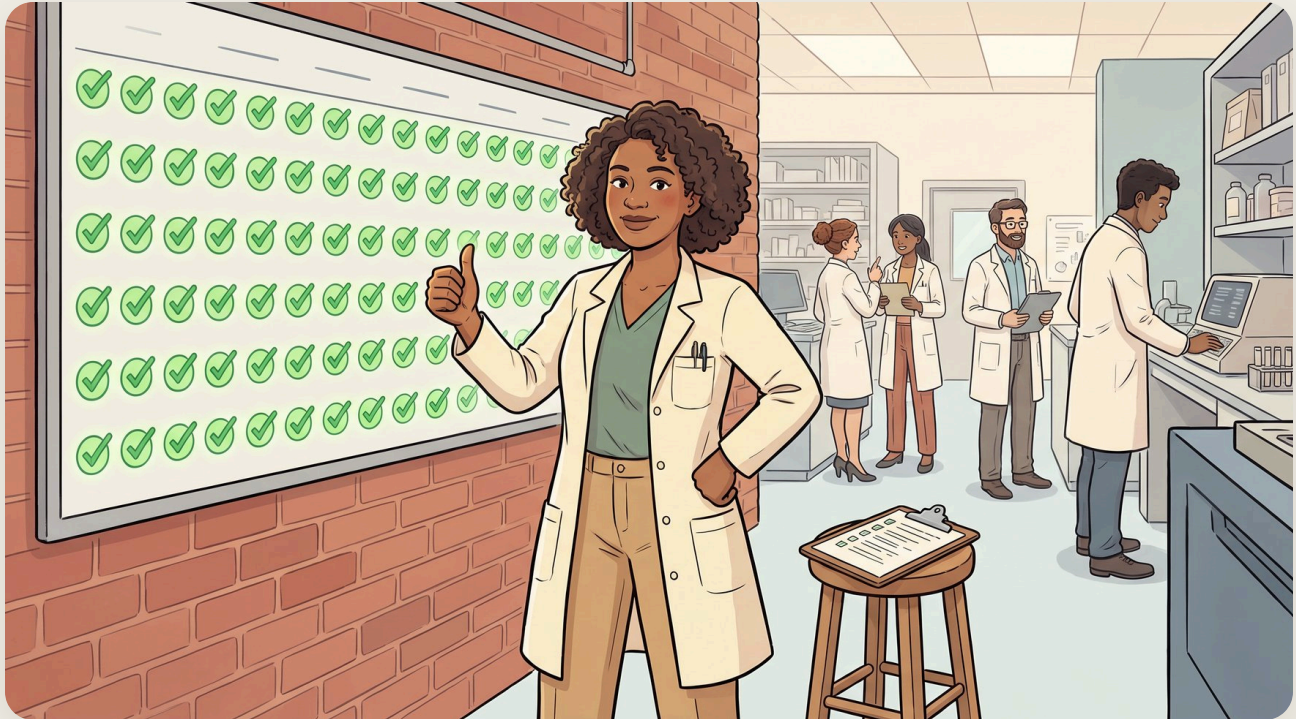
Config-gated and reversible

KEEPS

The estimator untouched

TESTING

Q8 How do we know it works?

**HOW BIRAGAS HELPS**

Thirty-four automated tests, seven smoke and twenty-seven integration, cover every module, the patch glue, the full config wiring, the node covariates, refutation, abstention and the end-to-end pipeline. Two real bugs were caught by these tests during the build: an invalid kernel-independence null and a silent orientation fallback, both fixed.

TESTS

34: 7 smoke + 27 integration

BUGS CAUGHT

2, both fixed

COVERS

Every module and the full pipeline

STATE

All passing

THE BENCHMARK**Q9** How well does it recover causal structure?**HOW BIRAGAS HELPS**

On a known ground-truth network the correlation rule scores F1 0.485 with 17 structural errors; the engine recovers the exact skeleton (F1 1.0, 2 errors), and a perfect causal graph with zero errors once one interventional anchor is supplied. Correlation, to exact skeleton, to perfect DAG. The pipeline also refutes spurious edges.

BASELINE

Correlation: F1 0.485, SHD 17

+ ONE ANCHOR

SHD 0 (perfect DAG)

ENGINE

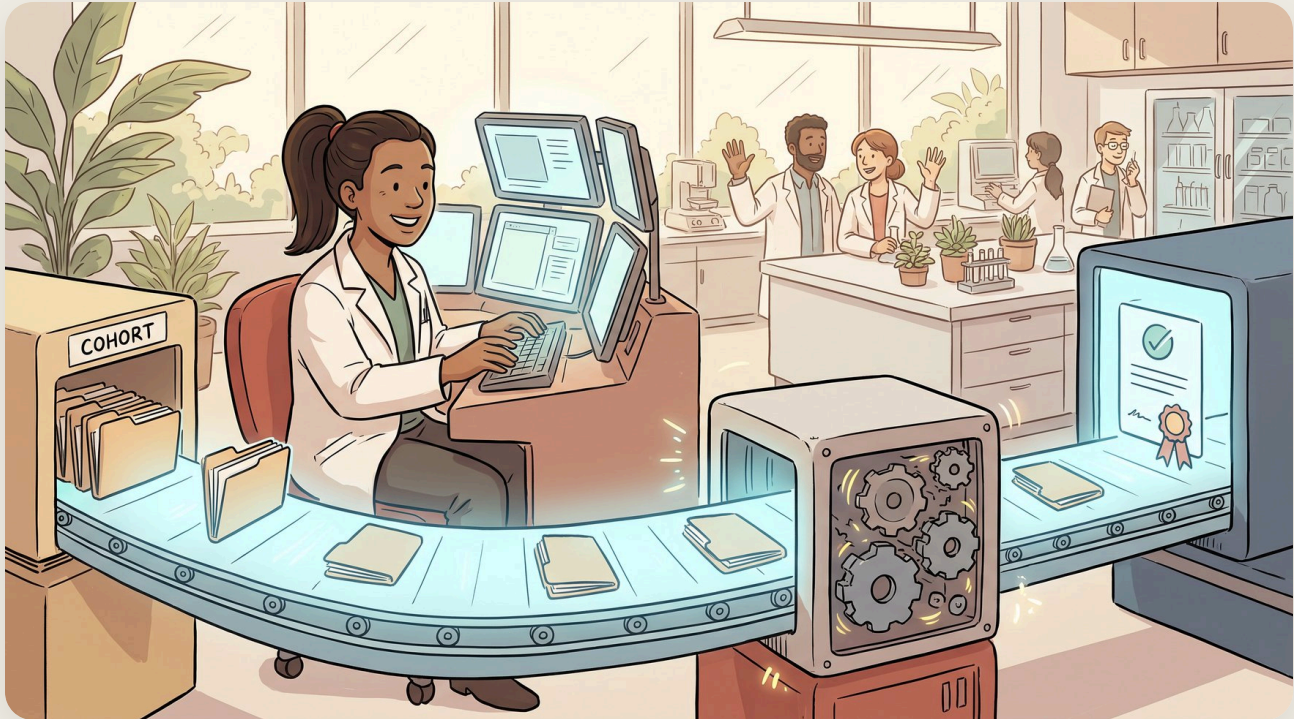
F1 1.0, SHD 2

ALSO

Refutes spurious edges

RUNNING IT

Q10 How do I run the full chain?



HOW BIRAGAS HELPS

Point the engine at your cohort expression and any module data, and one call runs discovery, model fitting, estimation, refutation and abstention to per-edge verdicts. Apply the two patches, set the config, populate the data keys, and validate with the tests and benchmark before a live run. Every claim traces to file and line; simulated data is labeled; nothing is fabricated.

ENTRY

`causal_pipeline(expression, evidence)`

OUTPUT

Per-target causal verdicts

STEPS

Patch, configure, populate, validate

INTEGRITY

File:line; nothing fabricated